Amit Kumar

Postdoctoral Researcher,
Bioinspired Soft Robotics Lab,
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Objective

Postdoctoral researcher developing adaptive biomaterials and environmentally responsive soft robotic systems through bioinspired material design, 4D printing, and stimuli-responsive polymers for biomedical and agricultural applications.

Research Area

My research area includes:

- **Functional Responsive Materials:** Hydrogel-, elastomer-, and nanoparticle-based systems with solvent and light programmable actuation.
- **Interfacial Polymer Chemistry:** Engineering dynamic biomaterials with biocompatible crosslinking strategies, bilayer architectures, and particle self-assembly to develop a novel multimaterial system for medical applications.
- **3D/4D Printing of Soft Materials:** Direct ink writing, bath-assisted printing, and molding of architectures that transform over time under environmental or external stimuli.
- **Biomedical Applications:** Designing biodegradable, biocompatible systems for autonomous controlled molecular delivery, adaptive antifouling property for targeted biomedical applications.

Education

- **Doctor of Philosophy (Ph.D.) + Master of Science (M.S.)- 9.75 (CGPA)** July'17 - July'24

  Dual Degree Interdisciplinary Research Program,
  Department of Biotechnology, Indian Institute of Technology Madras, Chennai, India

  Thesis: *Harnessing Solvent-Responsive Actuation Behavior of Polymers for Biomedical Applications*
  Supervisors: *Prof. Nitish R. Mahapatra, Department of Biotechnology,*
  Prof. Pijush Ghosh, Department of Applied Mechanics & Biomedical Engineering

- **Bachelor of Technology (B.Tech.)- 70.42 (CPI)** Aug'13 - May'17

  Department of Biotechnology, Delhi Technological University, Delhi, India

Professional Experience

- **Postdoctoral Researcher**, Istituto Italiano di Tecnologia (IIT), Italy 1st June'25 - Present

Mentor: Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics (BSR) Laboratory, IIT, Genoa

- **Hydrogel Powered Root Inspired Growing Robot (HyGROWBOT)**- Designing biodegradable hydrogel- elastomer composite actuators that can sense (pH, temperature) and move through the soil while enabling controlled release of bioactive molecules (nutrients, antimicrobial agents) in response to environmental cues- supporting sustainable soil health management.
- **Micropatterned surface for Superhydrophobic/Hydrophilic transition for Wound Dressing Application**- Developing multilayered chitosan-based wound dressing with waterproofing ability by changing from hydrophilic to superhydrophobic surface.
- **Sunlight-triggered Marangoni-powered Locomotion on Water**- Designing a multiresponsive actuator for on-demand environment-powered locomotion on the water.

- **Research Associate**, Indian Institute of Technology Madras (IITM), India Sept'24 - Feb'25

Mentor: Prof. Pijush Ghosh, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, IITM, Chennai

- **Industrial Trainee**, National Physical Laboratory (NPL), CSIR, Delhi, India June'16 - Aug'16

Mentor: Dr. G. Sumana Gajjala, Biomolecular Electronics and Conducting Polymer Research Group, NPL, CSIR

- **Industrial Trainee**, INMAS, DRDO, Delhi, India Dec'15 - Jan'16

Mentor: Dr. Aseem Bhatnagar, Department of Nuclear Medicine, INMAS, DRDO

Research Publications

1. **Amit Kumar**, Anas Saifi, Kumari Kiran, Sanjib Senapati, Nitish R. Mahapatra, Pijush Ghosh. Dual-Responsive Bidirectional Actuator for Biofilm Disruption on Medical Surfaces. *Advanced Healthcare Materials*, **(2026). (IF- 11.0)**
2. Anas Saifi, Smruti Parimita, **Amit Kumar**, Pijush Ghosh. Programmable 4D-Printed Soft Actuators: Harnessing Bending Strain Distribution for Embedded Topological Functionality. *ACS Applied Engineering Materials*, **(2025). (IF- 3.5)**
3. **Amit Kumar**[1], Shaik Sameer Basha[1], Mukilarasi B, Vimalraj Selvaraj, Pijush Ghosh, Swathi Sudhakar. Leveraging Cancer Microenvironment pH Shift for Actuation-induced Multidrug Release from Chitosan/Carboxymethyl Cellulose Bilayer Film. *International Journal of Biological Macromolecules*, **(2025). (IF- 8.7)**
4. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. 4D printing of pH-responsive bilayer with programmable shape-shifting behaviour. *European Polymer Journal*, **(2024). (IF- 6.8)**
5. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra, Pijush Ghosh. Superhydrophobic Asymmetric pH-responsive Soft Actuators: Implications for the Development of Anti-fouling Medical Devices. *Chemical Engineering Journal*, **(2024). (IF- 12.5)**
6. Nidhi Gupta[1], **Amit Kumar**[1], Pravin K Vaddavalli, Nitish R. Mahapatra, Akhil Varshney, Pijush Ghosh. Efficient Reduction of the Scrolling of Descemet Membrane Endothelial Keratoplasty Grafts by Engineering the Medium. *Acta Biomaterialia*, **(2023). (IF- 10.4)**
7. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. Solvent triggered shape morphism of 4D printed hydrogels. *Journal of Manufacturing Processes*, **(2023). (IF- 7.8)**
8. **Amit Kumar**, Raja Rajamanickam, Joyita Hazra, Nitish R. Mahapatra, Pijush Ghosh. Engineering the Nonmorphing Point of Actuation for Controlled Drug Release by Hydrogel Bilayer across the pH Spectrum. *ACS Applied Materials & Interfaces*, **(2022). (IF- 7.8)**

Research Publications (To Be Communicated/Under Review)

1. Ruowen Tu[1], **Amit Kumar**[1], Barbara Mazzolai. An Application-Driven Framework for Designing and Assessing Biodegradable Soft Robots. *Perspective- Soft Science, OAE Publishing-* **(Under Review)**
2. Yesheneh Jejaw Mamo, Ruowen Tu, **Amit Kumar**, Yasmin Tauqeer Ansari, Barbara Mazzolai, Nicola M. Pugno. Hierarchical Monolithic Yoshimura Origami with Hydrogel-Driven Programmable Shape Morphing for Untethered Soft Robotic Systems. *Advanced Intelligent Systems-* **(Under Review)**

Technical Skills

- **Laboratory Proficiencies:**
  - **Responsive biomaterials fabrication:** Hydrogels, Elastomers, Micro/Nano-particles, 3D printing (DIW)
  - **Advanced microscopy and spectroscopy:** Fourier Transform Infrared Spectroscopy (FTIR), Ultraviolet-Visible (UV-VIs) Spectroscopy, Scanning Electron Microscopy (SEM), Fluorescence Microscopy
  - **Cellular assays:** Live/dead bacterial viability, Antibacterial studies, Biocompatibility and Biodegradability studies
- **Computational and Analytical Tools:**
  - **Visualization & Design tools:** Origin, Graphpad Prism, Chemdraw, Inkscape, Blender, Filmora 9.0, Ultimaker Cura
  - **Documentation:** Microsoft Office, LaTeX

Posters and Presentations

1. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra and Pijush Ghosh,"Superhydrophobic Asymmetric Solvent Responsive Actuator for Antifouling Medical Devices", *Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza*, 5th July 2024, IIT Madras, Chennai, India.
2. **Amit Kumar**, Nidhi Gupta, Nitish R. Mahapatra and Pijush Ghosh, "Strategies to attain underwater actuation of solvent-responsive chitosan and its biomedical applications", *33rd Annual Conference of the European Society for Biomaterials* (ESB-2023), 4th – 8th September 2023, Davos, Switzerland.
3. **Amit Kumar**, V.P. Anju, Nitish R. Mahapatra and Pijush Ghosh, "Impact of Tuning Solvent Characteristics on Self-folding of Biopolymer Thin Film", *Active Polymeric Materials and Microsystems* (APMM-2019), 16th – 19th September 2019, Dresden, Germany.
4. **Amit Kumar**, Smruti Parimita, Nitish R. Mahapatra and Pijush Ghosh, "pH-Responsive Soft Actuator for Biomedical Applications", *3D Graphy Engineering and Medical 2023*, (3DGRAPHY- 2023), 9th - 10th December 2023, IIT Bombay, Maharashtra, India.
5. **Amit Kumar**, Nitish R. Mahapatra and Pijush Ghosh, "pH-responsive bilayer films with engineered non-morphing point of actuation for controlled model drug release", *14th International Conference on Advancements in Polymeric Materials* (APM-2023), 17th – 19th March 2023, CIPET (SARP)-APDDRL, India.

Workshop Attended

1. Tumor Microenvironment and Resistance to Drugs: The Role of Biomaterials, In Silico and In Vitro Modeling, 19th September 2025, University of Pavia, Italy.
2. Additive Manufacturing of Bio-Implants - Academic & Industry Perspectives, 10th - 11th March 2023, Indian Institute of Technology Madras, India.

Research Awards and Grants

- **Marie Curie Postdoctoral Fellowship- Seal of Excellence** 2026
  - My proposal "4DEX-mplant" secured 90.80% and was recognized as a high-quality project by the European Commission.
- **Research Excellence Award**- IIT Madras 2024
  - Best Research Publication in the Department of Biotechnology, awarded by IIT Madras during Dec 2023- Dec 2024.
- **Institute Research Award**- IIT Madras 2024
  - Best Doctoral Research in the Department of Biotechnology, awarded by IIT Madras.
- **First Prize**- Sci-Tech Exposium, IIT Madras 2024
  - First prize in Oral presentation at Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza.
- **Graduate Aptitude Test in Engineering (GATE)**- India 2017
  - National level scholarship exam to pursue M.S. (by research) from IIT Madras through HTRA scheme.

Position of Responsibility

- **Mentoring- PhD Scholar**, IIT Italy Dec'25 - Present
  - Provided PhD mentorship in designing hydrogel sacrificial mold-based 3D elastomer origami actuators and demonstrating reversible hydrogel swelling-driven actuation for programmable soft robotic systems.
- **Workshop Organiser- Ieee Ras Robosoft 2026**, Kanazawa, Japan April'26
  - Organized a full-day workshop titled "Functional and Untethered Soft Machines Powered by the Environment" with the theme "Can soft machines achieve true autonomy by dynamically adapting to environmental changes or by harvesting and utilizing ambient energy sources?". The workshop included: invited talks, a concept competition, and a poster competition.
- **Guest Editor**, Special Issue, Bioinspiration & Biomimetics, IOP Publishing Present
  - The Special Issue on the topic "Soft Robots and Actuators Powered by the Environment", seeks to consolidate and advance research at the intersection of soft robotics, bioinspiration, environmental energy harvesting, and sustainable design.
- **Teaching Assistant- Genetic Engineering Lab (BT2121)**, IIT Madras Aug'19 - Dec'19
  - Conducted competent cells, transformation, plasmid isolation, gel electrophoresis, restriction digestion, and PCR amplification experiments.
- **Teaching Assistant- Microbiology Lab (BT2112)**, IIT Madras Aug'18 - Dec'18
  - Successfully conducted experiments for media preparation, pure culture isolation, Gram staining, and selective media preparation.

Extracurricular Activities

- Secured 3rd place in both Football and Pole Vault (athletics) at Inter IIT Sports Meet, IIT Gandhinagar in December, 2023.
- Captained Institute Athletics team to secure Runner-Up position at Inter IIT Sports Meet, IIT Delhi in December, 2022.
- Held Inter IIT Meet Record for 3 editions (2018-2022) in Pole Vault with a height of 3.82m and won the Gold Medals for three consecutive years in Inter IIT Sports Meet among 21 IITs across India.

Academic Referees

1. Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics Lab, Italiano di Tecnologia (IIT), Genoa, Italy
2. Prof. Nitish R. Mahapatra, Professor, Cardiovascular Genetics Lab, Department of Biotechnology, Indian Institute of Technology Madras, India
3. Prof. Pijush Ghosh, Professor, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, Indian Institute of Technology Madras, India
4. Dr. Nidhi Gupta, Senior Consultant, Dr. Shroff's Charity Eye Hospital, India